Collateral Damage in Graph-based Memory Forgetting: Quantification, Exploitation, and Defense

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Abstract

Agent memory systems increasingly adopt entity graphs, and graph-based invalidation propagation—cascading updates through entity connections—is the natural next step for consistency maintenance. We show that this step is dangerous without safeguards: **structural co-occurrence does not imply dependency**, and naive BFS propagation over co-occurrence graphs damages 69–79% of valid memories across 6K–64K turns while degrading cross-task retrieval F1 by 10.2 ± 1.6pp. Under adversarial hub-targeting, five injected facts degrade 57.3% of valid memories—weaponizing the system's own maintenance mechanism. We propose **ATTR-AWARE Selective Propagation**, an attribute-keyed directional filter that restricts cascades to plausible dependency paths without LLM calls, achieving 78–100% damage reduction. Ablation confirms that directional filtering accounts for the entire damage reduction; in-degree dampening affects only severity. These findings establish that any system adopting graph-based forgetting over co-occurrence structures needs dependency-aware propagation from the outset.

1. Introduction

AI agents with persistent memory are moving from prototypes to production. Mem0 (Chhikara et al., 2025) and other memory-augmented agent frameworks have proliferated across open-source and proprietary ecosystems (Yang et al., 2026). These systems increasingly maintain entity graphs to organize stored facts (Chhikara et al., 2025; Jiang et al., 2026; Rasmussen et al., 2025).

As memory scales to tens of thousands of facts, maintaining consistency becomes critical—yet multi-hop conflict resolution accuracy remains below 6% on standard benchmarks (Hu et al., 2026). The gap is structural: current systems handle invalidation *individually*, leaving dependent facts silently stale. This creates concrete pressure toward propagation. Anecdotal reports from production deployments suggest that per-fact-only invalidation leads to accumulating inconsistencies over time.[1] Graph-based propagation—cascading updates through entity connections—is the natural next step, and the implementation distance is short. Zep (Rasmussen et al., 2025) maintains temporal validity windows on graph edges; MAGMA (Jiang et al., 2026) constructs explicit causal subgraphs; Mem0 (Chhikara et al., 2025) already traverses entity graphs during conflict detection. Adding weight updates to this existing traversal requires minimal code changes, creating strong pressure toward adoption. Each system implements prerequisites for propagation without taking the final step.

We implement this step and reveal a fundamental tension: **co-occurrence graphs conflate correlation with causation**. Facts sharing an entity are linked identically regardless of dependency—"the president of Russia" connects diplomatic and economic facts through the same edges. This is not an artifact of naive graph construction: even typed-relation graphs contain co-occurrence as an unavoidable substructure whenever multiple facts reference the same entity. BFS propagation treats all such connections identically, cascading invalidation through semantically meaningless edges.

Without this work, the first production system to adopt graph-based propagation over co-occurrence structures would discover the 69–79% damage rate in production—after corrupting its users' memories. This distinction is well-understood in transportation network resilience (Chen et al., 2007) but has not been applied to AI memory systems. We contribute:

1. **Quantified risk**: BFS propagation over co-occurrence graphs damages 69–79% of valid memories across

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[1]Mem0 GitHub Issues #4573, #4863, #3841 (2025) describe stale entity links and contradictory facts as recurring community complaints.

6K–64K scales and degrades cross-task retrieval F1 by 10.2 ± 1.6pp—turning a maintenance operation into a corruption vector (Section 4).

2. **Adversarial amplification**: Five injected facts exploit hub topology to degrade 57.3% of valid memories; the attack requires gray-box access (hub entity names) and weaponizes the system's own forgetting mechanism (Section 5).

3. **Practical defense**: ATTR-AWARE propagation, a hyperparameter-free attribute-keyed directional filter, reduces collateral damage by 78–100% without LLM calls. Ablation shows directional filtering alone accounts for the entire damage reduction (Section 6).

We characterize these failure modes *before* any production system adopts unfiltered graph propagation—providing quantified evidence and a working defense so that the first deployment does not have to discover the problem empirically.

2. Background

Graph-augmented memory systems. Modern AI memory systems differ in how they encode entity relationships. Table 1 summarizes existing architectures along the dimension most relevant to propagation: edge semantics and invalidation strategy. In their published descriptions, graphs serve *retrieval* and consolidation but none specifies a general-purpose *invalidation propagation* mechanism over entity co-occurrence edges.

System	Graph	Edges	Invalidation
Mem0[g]	KG	Typed	Per-fact
Zep	Temporal KG	Bi-temporal	Per-edge
MAGMA	4 sub-graphs	Causal+Entity	Per-graph
A-Mem	Linked notes	Associative	Per-note

Table 1. Graph-augmented memory systems (Chhikara et al., 2025; Rasmussen et al., 2025; Jiang et al., 2026; Xu et al., 2025). [g]graph-enabled mode. We found no explicit invalidation propagation in their published descriptions.

The forgetting gap. MemoryAgentBench (Hu et al., 2026) evaluates multi-hop conflict resolution where systems must update or discard outdated facts under contradictory evidence. No tested system exceeds 6% accuracy. MemoryArena (He et al., 2026) shows agents drop to 40–60% on session-interdependent tasks. The failure mode is consistent: systems cannot reason about *dependencies between facts*.

Belief revision perspective. Kumiho (Park, 2026) formalizes this problem through AGM belief revision (Alchourrón et al., 1985): typed dependency edges (e.g., `Depends_On`, `Derived_From`) ensure that only causally related beliefs are affected during revision. This provides axiomatic guarantees—but requires typed edges at graph construction time, which Mem0, Zep, and MAGMA do not currently provide. Our work addresses the complementary question: *what happens when propagation is attempted on existing untyped co-occurrence graphs?*

Memory as an attack surface. MINJA (Dong et al., 2025) achieves 98% injection success via query-only interaction; AgentPoison (Chen et al., 2024) poisons 80%+ of agent decisions by modifying <0.1% of stored entries. Xu et al. (2026) show that memory retrieval can steer agent control flow, while Devarangadi Sunil et al. (2026) evaluate poisoning defenses under realistic conditions with pre-existing legitimate memories. Xiong et al. (2025) empirically demonstrate that memory addition/deletion policies significantly impact long-term agent performance. MITRE ATLAS designates memory manipulation as technique AML.T0080 (MITRE Corporation, 2023). These attacks inject malicious *content*; our work reveals a structurally distinct threat where the system's own *maintenance mechanism* becomes the attack vector.

Structural vulnerability from graph topology. The content-injection attacks above exploit *what* is stored; a complementary question is whether the graph *structure itself* creates vulnerability. Networks with heavy-tailed degree distributions are robust to random failures but catastrophically fragile to targeted hub removal (Albert et al., 2000; Cohen et al., 2000). Transportation network resilience provides a structural analogy: directional failure models separate correlated from causally dependent links (Chen et al., 2007), though the dynamics differ between physical and information networks. We test whether memory co-occurrence graphs exhibit this topology and whether hub-targeted operations produce the predicted damage.

3. The Propagation Trade-off

Before analyzing failure modes, we establish *why* propagation matters. We design 10 fact-change scenarios (personnel changes, schedule updates) with 5 causally related and 15 unrelated memories each. For instance, updating "the CEO of Acme is Alice" to "the CEO of Acme is Bob" should invalidate "Alice approves Acme milestones" (dependent) but not "Acme was founded in 2010" (co-occurring but independent).

Finding 1. Table 2 shows that No-Propagation leaves substantial stale content (16% average, 40% worst case). BFS eliminates staleness but damages 47.5% of unrelated memories—producing more damage than the staleness it eliminates. ATTR-AWARE achieves near-BFS forgetting

Mode	Forget Acc.	Collateral	Stale@worst
No-Propagation	84.0%	0.0%	40%
BFS	96.0%	47.5%	0%
ATTR-AWARE	94.0%	25.0%	8%

Table 2. Forgetting benefit vs. collateral cost (10 controlled scenarios). No-Propagation leaves 16% stale (worst 40%). BFS achieves highest accuracy but damages 47.5% of unrelated memories. ATTR-AWARE reduces collateral, though 25% residual indicates room for improvement.

accuracy (94% vs 96%) while reducing collateral damage, demonstrating that *if propagation is adopted*, dependency-aware filtering is essential.

The remainder of this paper characterizes the failure modes that make unfiltered propagation dangerous (Sections 4 and 5) and validates the defense (Section 6).

4. Collateral Damage at Scale

4.1. Setup

We run FactConsolidation memorization on MemoryAgentBench (Hu et al., 2026) under three modes: Baseline (no propagation), BFS (BFS through co-occurrence graph), and ATTR-AWARE (attribute-directed chains only). Infrastructure: Gemma-4-31B-it (LLM), BAAI/bge-m3 1024-dim (embedding), NetworkX graph, depth=2, decay_per_hop=0.5. No current system deploys BFS propagation; we implement it because existing graph infrastructure makes adoption straightforward, and characterizing its failure modes before deployment is the point of proactive safety research.

We define five metrics: *damage*—any memory weight reduction ($w < 1.0$) from propagation; *severe damage*—weight reduced to retrieval-degrading levels ($0.1 \leq w < 0.5$); *kill*—weight below retrieval threshold ($w < 0.1$); *stale retention*—outdated facts that should have been invalidated but remain unchanged; *retrieval degradation*—Exact Match (EM) or F1 decrease on unrelated downstream tasks.

4.2. Scale Trend Results

Scale	Valid	BFS Dam.	Attr Dam.	Def.	Hub$_{\max}$
6K	344	68.9%	14.8%	78.5%	23
32K	1,744	79.0%	15.8%	80.0%	121
64K	3,414	78.5%	15.0%	80.9%	298

Table 3. Collateral damage across scales (MemoryAgentBench FactConsolidation). BFS damage increases then stabilizes; ATTR-AWARE stays at ∼15%. Hub$_{\max}$: max entity degree. Def. = $1 - \text{Attr}/\text{BFS}$.

Finding 2. BFS damage ranges from 69–79% across scales (Table 3). Hub degree grows $\sim 13\times$ across scales versus $\sim 10\times$ fact growth, amplifying vulnerability.

Finding 3. ATTR-AWARE damage remains stable at ∼15% regardless of scale. Full classification of all 51 organically damaged memories at 6K found 88.2% legitimate cascade, 11.8% borderline, and zero false positives (see Limitation 3), confirming that this residual reflects conservative dependency-following rather than error.

4.3. Graph Topology

The degree distributions (Figure 1) are **heavy-tailed** across all scales ($\alpha \in [2.73, 2.96]$ via MLE power-law fitting (Clauset et al., 2009)), though the strict power-law hypothesis is rejected at 64K in favor of log-normal. The key driver of propagation damage is degree heterogeneity $\kappa \equiv \langle k^2 \rangle / \langle k \rangle$, which grows from 4.6 to 32.3 across scales—well above the Molloy-Reed threshold ($\kappa > 2$) that predicts catastrophic fragmentation under targeted hub removal (Molloy & Reed, 1995; Albert et al., 2000; Cohen et al., 2000).

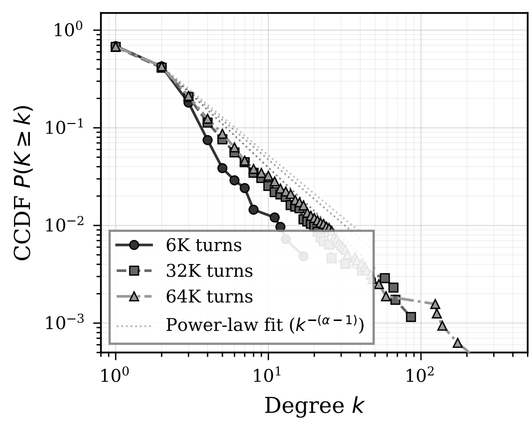

Figure 1. Complementary cumulative degree distribution (CCDF) on log-log axes. Dotted lines: MLE power-law fits. The heavy tail extends with scale.

Graph construction control. A natural concern is whether co-occurrence clique construction inflates hub degrees. We compare against a *typed-relation* graph at 6K that retains only subject→object edges (no cliques): edge count decreases by 10.4%, but BFS damage drops only 1.4pp (38.4% vs 39.8%). This suggests that hub degrees are driven primarily by Zipfian entity frequency, not clique construction artifacts. Extension to 32K/64K remains for future work (see Limitations).

4.4. Cross-task Contamination

BFS propagation during FactConsolidation degrades Accurate_Retrieval F1 by 10.2 ± 1.6pp across the top-5 hub entities (Table 9); the worst hub (Debbie, degree 715) cascades through 3,381 memories and causes -11.9pp F1 on completely unrelated queries. Damage correlates strongly with hub degree ($r = -0.98$). ATTR-AWARE preserves accuracy entirely (ΔF1 = $+0.1$pp), confirming that directional filtering prevents cross-task interference. This is the most direct evidence that structural damage translates to retrieval degradation, and motivates a follow-up question: could hub-concentrated damage be deliberately triggered?

5. Hub-Targeted Amplification

5.1. Threat Model

We evaluate three attacker knowledge levels: **black-box** (general knowledge), **gray-box** (hub entity names known), and **white-box** (fact registry access). The attacker injects facts through the system's normal memory write interface; conflict detection triggers when a new fact maps to an existing subject-key, initiating the propagation cascade under study. This write path is realistic: Mem0 and Zep expose memory-write APIs, and MINJA (Dong et al., 2025) demonstrates that even query-only interaction can induce memory writes through indirect prompt injection. We analyze this vector to inform defensive design; responsible evaluation of adversarial risks is essential for proactive safety.

5.2. Results

Attack	BFS Dam.	BFS Kill	Attr Dam.
6K Context (344 valid memories)			
1 fact	5.2%	0.3%	0.0%
3 facts	26.7%	4.4%	0.0%
5 facts	39.8%	17.4%	0.3%
32K Context (1,744 valid memories)			
1 fact	38.1%	1.0%	0.0%
5 facts	**57.3%**	**19.3%**	1.0%

Table 4. White-box adversarial injection at 6K and 32K scales. Kill: weight reduced below retrieval threshold (< 0.1). Five facts at 32K damage 57.3% of valid memories.

Finding 4. A single fake fact damages 5.2% (6K) to 38.1% (32K) of valid memories (Table 4)—7.3× amplification from scale and hub-degree growth. Five facts at 32K degrade 57.3%, with 19.3% killed (weight < 0.1).

Attack	Facts	Hit	BFS Dam.
White-box	5	100%	39.8%
Gray-box	5	100%	20.9%
Black-box	5	0%	0.0%

Table 5. Attack comparison at 6K. Black-box fails entirely; gray-box achieves 52% of white-box damage.

Finding 5. Table 5 shows that under our random-entity black-box baseline, attacks fail to hit hub entities (0% hit rate). At minimum, gray-box knowledge of graph structure is needed for the attack vector studied here. Figure 2 shows that damage grows sublinearly with propagation depth while kill rate accelerates due to cumulative decay.

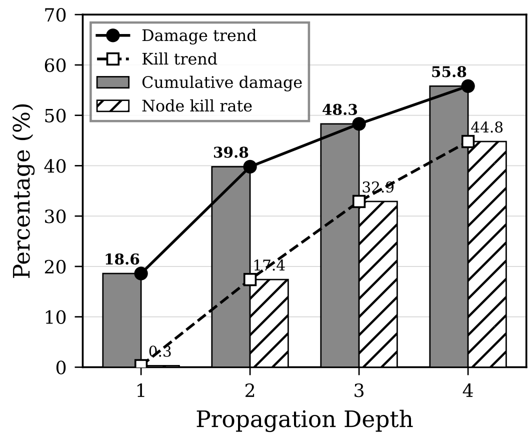

Figure 2. Propagation depth vs collateral damage (6K, 5-fact attack). Damage grows sublinearly while kill rate accelerates due to cumulative decay.

6. ATTR-AWARE Selective Propagation

6.1. Method

The core intuition is directional filtering. Consider updating "the CEO of Acme is Alice" to "…is Bob." In this fact, *Acme* is the subject and *Alice* is the object. Facts *about Alice* (where Alice is subject)—e.g., "Alice approves Acme milestones"—may need updating because they depend on Alice's role. Facts *about Acme* that merely co-occur with Alice—e.g., "Acme was founded in 2010"—do not, because Acme's founding is independent of who its CEO is. ATTR-AWARE operationalizes this by following only object→subject chains: when a fact with subject s and object o is invalidated, propagation continues only to facts where o appears as *subject* (line 4 in Algorithm 1)—restricting cascades to plausible dependency paths:

The key filtering step is line 4: instead of visiting all graph neighbors, ATTR-AWARE retrieves only facts where the object entity appears as *subject*—restricting propagation to downstream attribute-directed chains. ATTR-AWARE does not infer causality directly; it uses attribute-keyed directionality as a conservative proxy for dependency. The in-degree adjustment (line 6) dampens decay for highly-connected entities. BFS differs by visiting all graph neighbors without semantic filtering; both methods apply in-degree dampening $(1 - (1 - \delta^h)/d_{\text{in}})$ to reduce penalty for highly-connected entities. All operations are string-matching or arithmetic over pre-existing metadata—**no LLM calls** at propagation time. At 6K scale (417 nodes, 524 edges), a 5-fact attack

Algorithm 1 ATTR-AWARE Selective Propagation

Require: Invalidated fact f with subject s, graph G, registry R, max depth D, decay δ, current depth $h=0$

$s \leftarrow$ subject entity of f
$E_{\text{obj}} \leftarrow$ entities linked to f in G, $e \neq s$
for each $e \in E_{\text{obj}}$ **do**
  $M_{\text{down}} \leftarrow \{R[k] \mid k.\text{entity} = e\}$ {facts where e is subject}
  for each $m \in M_{\text{down}}$ if m is valid **do**
    $d_{\text{in}} \leftarrow \max(\text{in\_degree}(e, G), 1)$
    $m.w \leftarrow m.w \times (1 - \frac{1-\delta^{h+1}}{d_{\text{in}}})$
    if $h < D$ **then**
      Propagate$(m, G, R, D, \delta, h{+}1)$
    end if
  end for
end for

completes propagation in 2.3 ± 0.2 ms for ATTR-AWARE versus 12.8 ± 1.2 ms for BFS—a $5.5\times$ speedup from reduced traversal scope. Propagation constitutes <5% of total processing time (dominated by LLM conflict detection), and the full pipeline processes 64K-scale graphs in <1 s without GPU.[2]

6.2. Propagation Strategy Comparison

We compare nine propagation strategies spanning structural, weight-based, and semantic filtering approaches (Table 6).

Strategy	Dam.	Kill	EM	F1
Unfiltered co-occurrence traversal				
BFS (standard)	39.8%	17.4%	18.0%	18.3%
BFS + visited set	39.8%	10.5%	15.0%	15.3%
Structural constraints				
Degree-cap=5	24.1%	8.1%	19.0%	19.3%
Degree-cap=10	28.2%	10.8%	18.0%	18.3%
Threshold-based filtering				
k-hop thresh. θ=0.3	18.6%	0.3%	21.0%	21.0%
k-hop thresh. θ=0.5	18.6%	0.3%	21.0%	21.0%
Weighted BFS θ=0.1	11.3%	3.2%	21.0%	21.3%
Weighted BFS θ=0.3	1.5%	0.0%	18.0%	18.3%
Semantic filtering				
ATTR-AWARE	**0.3%**	**0.0%**	17.0%	17.3%

Table 6. Propagation strategy comparison (6K, 5-fact white-box attack). EM/F1 evaluated on n=100 adversarial queries; differences are not statistically significant (all strategies fall within 17–21%). Damage and kill metrics are deterministic.

Finding 6. Table 6 reveals a damage spectrum from structural constraints (degree capping, modest reduction) through threshold-based filters (weighted BFS at θ=0.3: 1.5% damage) to semantic filtering (ATTR-AWARE: 0.3%). Threshold methods require per-deployment tuning and exhibit cliff effects (k-hop θ=0.3 and 0.5 produce identical results); ATTR-AWARE requires no hyperparameters. At 32K/64K scale, Weighted BFS θ=0.3 maintains 0.2%/0.1% damage—comparable to ATTR-AWARE's 1.0%/0.9%—while BFS standard damage escalates to 57.3%/61.6% (from 39.8% at 6K). Both filtered strategies remain effective even as hub degrees grow to 298; the practical advantage of ATTR-AWARE lies in hyperparameter-free operation, avoiding the cliff effects and per-deployment θ-tuning that threshold methods require. EM/F1 scores (n=100) converge to 17–21% across all strategies—the adversarial query set (5 injected facts, 100 questions) lacks sensitivity to discriminate what damage metrics clearly separate. The robust downstream signal is cross-task contamination (Appendix A): BFS degrades F1 by 10.2 ± 1.6pp on *unrelated* queries; ATTR-AWARE preserves accuracy entirely.

Ablation: direction vs. dampening. ATTR-AWARE combines two mechanisms: (i) directional filtering (object→subject chains) and (ii) in-degree dampening $(1 - (1-\delta^h)/d_{\text{in}})$. To isolate their contributions, we run four variants at 6K (5-fact white-box attack). Removing dampening from BFS leaves damage unchanged (39.8%) but doubles kill rate (17.4%→33.7%); removing dampening from ATTR-AWARE has no effect (0.3% damage, 0% kill in both cases). **The entire 39.5pp damage reduction is attributable to directional filtering; dampening affects only the severity distribution, not the reach of propagation.** To test whether object→subject is the right direction, we compare three filtered variants: *downstream* (object→subject, the default), *upstream* (subject→object), and *bidirectional*. Downstream achieves the lowest damage (0.3%, 1 memory); upstream damages 1.2% (4 memories) by propagating to other attributes of the same subject entity; bidirectional damages 1.5% (5 memories). All three reduce damage by ≥96% relative to BFS—confirming that *any* directional filter dramatically outperforms undirected traversal—but downstream is strictly optimal, consistent with the dependency-follows-object intuition.

6.3. Cross-experiment Consistency

Finding 7. ATTR-AWARE achieves 78–100% defense effectiveness across all conditions (Table 7). The strongest signal is cross-task: ATTR-AWARE fully preserves retrieval accuracy (ΔF1 = 0.0pp) where BFS degrades it by 10.2 ± 1.6pp.

[2]The 64K timing is based on end-to-end pipeline observation; controlled wallclock measurements at 32K/64K scale remain for future work. The 6K micro-benchmark provides the only formally measured propagation latency.

Experiment	BFS	WBFS	Attr	Def.
Scale 6K–64K	69–79%	—	15%	78–81%
Cross-task (F1)	−10.2pp	—	+0.0pp	100%
Adv 6K / 5f	39.8%	1.5%	0.3%	99.3%
Adv 32K / 5f	57.3%	0.2%	1.0%	98.2%
Adv 64K / 5f	61.6%	0.1%	0.9%	98.5%

Table 7. Defense effectiveness across all experiments. **Def.** = $1 - \text{Attr}/\text{BFS}$. WBFS = Weighted BFS θ=0.3 (best per-scale). At 32K/64K, WBFS achieves lower absolute damage than ATTR-AWARE, but requires per-deployment θ-tuning. Cross-task: BFS degrades F1 by 10.2 ± 1.6pp; ATTR-AWARE preserves accuracy.

6.4. Robustness to BFS Variants

A natural concern is whether BFS damage is inflated by redundant decay (a node decayed multiple times via different paths) or a particular decay parameter. We test variants at both 6K and 32K under the 5-fact white-box attack (Table 8).

Scale	BFS Variant	Dam.	Sev.	Kill
6K	Standard (decay=0.5)	39.8%	11.1%	17.4%
	+ memory visited set	39.8%	—	10.5%
	decay = 0.3	39.8%	11.9%	20.1%
	decay = 0.7	39.8%	19.8%	2.0%
	decay = 0.9	39.8%	2.0%	0.0%
32K	Standard (decay=0.5)	57.3%	—	19.3%
	+ memory visited set	57.3%	—	7.8%

Table 8. BFS damage is structurally determined: damage is identical across all variants within each scale (39.8% at 6K, 57.3% at 32K). Sev. = severe damage ($0.1 \leq w < 0.5$). At the default decay=0.5, 28.5% of valid memories suffer kill or severe damage.

Finding 8. Table 8 reveals that damage is invariant to both redundant-decay elimination and decay parameter choice at *both* scales. The 39.8% (6K) and 57.3% (32K) damage reflects BFS *reachability*—the fraction of memories structurally connected to hub entities within depth 2—not an artifact of parameter tuning or implementation choices. At the default decay of 0.5, 28.5% of valid memories suffer severe or kill-level weight reduction ($w < 0.5$), confirming that damage is not merely cosmetic. Even at decay=0.9 (the most conservative setting), 2.0% of memories are severely damaged despite zero kills—and the 39.8% damage persists. The higher 32K rate confirms that structural vulnerability increases with scale as hub degree grows ($23 \rightarrow 121$).

7. Related Work

Memory invalidation and belief revision. Kumiho (Park, 2026) maps AGM belief revision postulates (Alchourrón et al., 1985) to a property graph with typed dependency edges, providing formal guarantees that unrelated beliefs are preserved. Our work is complementary: Kumiho shows *why* typed edges are needed (axiomatic); we show *what happens without them* (empirical), providing the damage quantification that motivates architectural choices like Kumiho's. ATTR-AWARE's object→subject chain is structurally analogous to justification chains in Truth Maintenance Systems (Doyle, 1979), but implemented via lightweight string-matching over pre-existing metadata rather than full dependency tracking. MaRS (Alqithami, 2025) compares six forgetting policies with provenance closure constraints on the retention side; our study addresses the invalidation side.

Memory security. MINJA (Dong et al., 2025), AgentPoison (Chen et al., 2024), and ER-MIA (Piehl et al., 2026) inject malicious content. KEPo (Chen et al., 2026) and LogicPoison (Xiao et al., 2026) manipulate graph topology. GraphRAG (Edge et al., 2024) uses community-structured graphs for retrieval; its invalidation under entity updates has not been studied. Our attack is structurally distinct: it triggers the system's own forgetting mechanism rather than injecting content or manipulating retrieval paths.

Network resilience. Heavy-tailed networks exhibit robust-yet-fragile behavior under targeted hub removal (Albert et al., 2000; Cohen et al., 2000). Our results are consistent with this pattern in AI memory graphs ($\alpha \approx 2.7$–3.0, heavy-tailed).

8. Discussion and Conclusion

Key message. Unfiltered co-occurrence propagation is a latent risk in graph-based memory systems. It damages 69–79% of valid memories, enables low-cost adversarial amplification (5 facts $\rightarrow$ 57.3% degradation), and contaminates unrelated tasks (F1 -10.2pp). ATTR-AWARE and Weighted BFS θ=0.3 each reduce these risks by ≥98%—the former without hyperparameters, the latter with lower absolute damage at scale but requiring per-deployment tuning. Ablation confirms that directional filtering drives the entire damage reduction. The design principle is clear: **any system propagating invalidation over co-occurrence structures needs dependency-aware filtering from the outset**.

Implications. Hub entities are essential for retrieval and cannot be removed, making them permanent structural vulnerabilities under unfiltered propagation. ATTR-AWARE functions as a safety guardrail that constrains propagation—including future learned policies (Alqithami, 2025)—to plausible dependency chains. Entity mention frequencies follow Zipf's law (Zipf, 1949), producing hub-and-spoke structures wherever entities are referenced with natural frequency (Steyvers & Tenenbaum, 2005; Barabási & Albert, 1999); the structural vulnerability we quantify is not benchmark-specific but a consequence of this universal pattern.

Limitations and future work. (1) Single benchmark and single LLM: while entity extraction is rule-based and graph topology drives core findings, replication across benchmarks and model families (e.g., GPT-4o, Llama-3) is needed. (2) ATTR-AWARE's accuracy at 6K: manual inspection found 88.2% legitimate cascade, 11.8% borderline, zero false positives among 51 organic-damaged memories; the false negative rate (missed dependencies) remains unmeasured. The 69–79% BFS damage is an upper bound on collateral—some fraction may reflect genuine dependencies. (3) Propagation depth fixed at 2; clique construction modestly inflates hub degree (1.4pp effect at 6K); both warrant multi-scale analysis. (4) Our direction ablation confirms object→subject is optimal among the three structural variants tested; embedding-similarity-based filters and validation on production systems (Mem0^g, Zep) once they adopt propagation are important next steps.

LLM/Agent Usage Disclosure

This research used LLM-based tools in three capacities: (1) Literature search and survey assistance (paper discovery and classification); (2) LaTeX editing and proofreading; (3) Experimental code development (benchmark integration, graph analysis scripts). All experimental results, analysis, and conclusions were produced and validated by the author. The core experimental pipeline (entity extraction, subject-key matching, propagation algorithms) uses rule-based methods without LLM involvement at runtime, except for conflict detection which uses Gemma-4-31B-it as specified in the experimental setup.

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A. Cross-task Contamination (C2)

We memorize 64K EventQA turns, then trigger propagation from each of the top-5 hub entities in isolation (snapshot-restore between hubs) and evaluate Accurate_Retrieval queries (n=100) that are *unrelated* to the propagation trigger. Baseline F1 = 70.9%.

Hub	**Deg.**	**Aff.**	**BFS F1**	**ΔF1**	**Attr F1**
Debbie	715	3,381	59.0%	−11.9pp	70.9%
And	643	3,424	59.2%	−11.8pp	70.9%
She	627	2,587	61.9%	−9.1pp	69.5%
Marianne	589	3,064	61.0%	−10.0pp	73.0%
But	571	3,342	62.8%	−8.2pp	70.9%
Mean±SD	629	3,160	60.8%	−10.2±1.6pp	71.1%

Table 9. Per-hub cross-task contamination (EventQA 64K → Accurate_Retrieval). **Aff.**: memories affected by BFS propagation from each hub. **ΔF1**: BFS F1 minus baseline. ATTR-AWARE preserves accuracy (ΔF1 = +0.1±1.3pp, within noise).

BFS degrades F1 by 10.2 ± 1.6pp across all five hubs, with the worst hub (Debbie, degree 715) cascading to 3,381 memories and causing -11.9pp F1 degradation on completely unrelated queries. Damage scales monotonically with hub degree ($r = -0.98$ between degree and ΔF1). ATTR-AWARE preserves full accuracy: mean F1 = $71.1 \pm 1.3\%$ versus baseline 70.9%, confirming that directional filtering prevents cross-task interference entirely.

B. Multi-run Statistics

We run the 6K adversarial attack 7 times (19 paired observations). Pooled Wilcoxon signed-rank test: $W = 86.0$, $p = 0.0022$; bootstrap 95% BCa CI $[2.82, 15.67]$pp excludes zero. A mixed-effects model with attack strength as fixed effect and run as random intercept confirms the 5-fact condition is significant ($p = 0.020$).

C. Graph Construction

Entity extraction uses rule-based proper noun detection and "the X of Y" patterns. Co-occurrence edges form cliques per fact. Subject-keys are deterministic: "the president of Russia is Putin" → `Russia:president`. Conflict is triggered when a new fact maps to an existing subject-key.

D. Experimental Configuration

LLM: Gemma-4-31B-it (temp=0.0). Embedding: BAAI/bge-m3 (1024-dim). Graph: NetworkX DiGraph. Storage: SQLite in-memory. Benchmark: MemoryAgent-Bench (ICLR 2026). Scales: 6K/32K/64K. Propagation: depth=2, decay=0.5.